

various purposes, and the BOLO will provide a useful data point for market participants to consider in addition to the NBBO.

5. Compliance Costs

Various market participants will incur one-time implementation costs as well as ongoing compliance costs to comply with the Rule. These costs and their computations are discussed in greater detail below, but are summarized in Table 15. Some of the costs are associated with the acceleration of aspects of the MDI Rules and will only represent new costs (which are not already anticipated under the MDI rules) if the exclusive SIPs do not become competing consolidators once the MDI rules are fully implemented.

Table 15: Compliance Cost Estimates						
Rule	Affected entities	One-time costs	Ongoing Costs	# of entities	Total one-time costs	Total ongoing costs
612	All trading venues ^a	\$156,000		277	\$43,212,000	
612	Listing exchanges ^b	\$33,000	\$9,000	5	\$165,000	\$45,000
612	SIPs ^c	\$13,000	\$9,000	2	\$26,000	\$18,000
612	Broker-dealers with order entry systems ^d	\$33,000		1,161	\$38,313,000	
612	Broker-dealers with smart order routers ^e	\$11,000		270	\$2,970,000	
610	Exchanges ^f	\$57,000		15	\$855,000	
600,603	Exchanges ^g	\$3,500	\$6,500	16	\$56,000	\$104,000
600,603, 612	SIPs ^h	\$613,000	\$174,000	2	\$1,226,000	\$348,000
	Total				\$86,823,000	\$515,000
Sources: Across estimates below, salaries are derived from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits and overhead. The burden hours estimates are based on Commission's experiences with burden estimates.						
^a See Proposing Release, <u>supra</u> note Error! Bookmark not defined. , at 80333. The Proposing Release's estimate of \$140,000 is adjusted to \$156,000 to account for a 11.4% increase in the Producer Price Index for Data Processing, Hosting and Related Services from December 2014, when the \$140,000 estimate was first made. See						

U.S. Bureau of Labor Statistics, Producer Price Index by Industry: Data Processing, Hosting and Related Services: Hosting, Active Server Pages (ASP), and Other Information Technology (IT) Infrastructure Provisioning Services [PCU5182105182105], retrieved from FRED, Federal Reserve Bank of St. Louis; available at <https://fred.stlouisfed.org/series/PCU5182105182105> (Mar. 11, 2024).

^b The \$33,000 estimate per listing exchange is based on the following calculations: \$19,950 (hourly rate for Sr. Programmer at \$399 for 50 hours) + \$6,860 (hourly rate for Sr. Systems Analyst at \$343 for 20 hours) + \$3,730 (hourly rate for Compliance Manager at \$373 for 10 hours) + \$2,940 (hourly rate for Director of Compliance at \$588 for 5 hour) $\approx$ \$33,000, for a total annual monetized burden of \$165,000 (i.e., \$165,000 = \$33,000 x 5 listing exchanges). The \$9,000 estimate per listing exchange is based on the following calculations: (\$2,640 (hourly rate for Compliance Attorney at \$440 for 6 hours) + \$746 (hourly rate for Compliance Manager at \$373 for 2 hours)) x 4 tick size revisions per year $\approx$ \$9,000, for a total annual monetized burden of \$45,000 (i.e., \$45,000 = \$9,000 x 5 listing exchanges).

^c The \$13,000 estimate per listing exchange is based on the following calculations: \$4,788 (hourly rate for Sr. Programmer at \$399 for 12 hours) + \$1,715 (hourly rate for Sr. Systems Analyst at \$343 for 5 hours) + \$3,730 (hourly rate for Compliance Manager at \$373 for 10 hours) + \$2,940 (hourly rate for Director of Compliance at \$588 for 5 hour) $\approx$ \$13,000. The \$9,000 estimate per listing exchange is based on the following calculations: (\$2,640 (hourly rate for Compliance Attorney at \$440 for 6 hours) + \$746 (hourly rate for Compliance Manager at \$373 for 2 hours)) x 4 tick size revisions per year $\approx$ \$9,000.

^d The \$11,000 estimate per system change is based on the following calculations: (\$2,005 (hourly rate for Attorney at \$401 for 5 hours) + \$2,980 (hourly rate for Compliance Manager at \$298 for 10 hours) + \$4,640 (hourly rate for Programmer Analyst at \$232 for 20 hours) + \$1,325 (hourly rate for Senior Business Analyst at \$265 for 5 hours) $\approx$ \$11,000. The Commission expects that broker-dealers are likely to have to undertake 3 system changes, for a total one-time expense of \$33,000. See also Transaction Fee Pilot Adopting Release, infra note 1657, at 5271 n.770.

^e The \$11,000 estimate per broker-dealers with smart order routers is based on the following Manager at \$298 for 10 hours) + \$4,640 (hourly rate for Programmer Analyst at \$232 for 20 hours) + \$1,325 (hourly rate for Senior Business Analyst at \$265 for 5 hours) $\approx$ \$11,000. See also Transaction Fee Pilot Adopting Release, infra note 1657, at 5274 n.796 where the cost to broker-dealers to update systems for the TSP was estimated to be \$9,000. Here, we are allowing for an additional 10 hours of Programmer Analyst time.

^f See Proposing Release, supra note **Error! Bookmark not defined.**, at 80333.

^g The additional \$3,500 in one-time costs and \$6,500 in ongoing costs represent a 5% addition over the costs reported in the MDI release. See supra note **Error! Bookmark not defined.**, section V.C.2(d)(ii) to account for the new requirement to send the necessary data to generate odd-lot information to the exclusive SIPs.

^h The \$613,000 estimate in one-time costs is based on the following calculations: \$33,000 (costs under the amendments to Rule 612 to update data specifications and internally and externally test the updates) + \$167,670 (\$83,790 (hourly rate for Sr. Programmer at \$399 for 210 hours) + \$61,740 (hourly rate for Sr. Systems Analyst at \$343 for 180 hours) + \$7,460 (hourly rate for Compliance Manager at \$373 for 20 hours) + \$5,880 (hourly rate for Director of Compliance at \$588 for 10 hours) + \$8,800 (hourly rate for Compliance Attorney at \$440 for 20 hours) + \$412,500 (costs for external services). The \$174,000 estimate in ongoing costs is based on the following calculations: \$50,301 (\$25,137 (hourly rate for Sr. Programmer at \$399 for 63 hours) + \$18,522 (hourly rate for Sr. Systems Analyst at \$343 for 54 hours) + \$2,238 (hourly rate for Compliance Manager at \$373 for 6 hours) + \$1,764 (hourly rate for Director of Compliance at \$588 for 3 hours) + \$2,640 (hourly rate for Compliance Attorney at \$440 for 6 hours)) + \$123,725 (costs for external services). See infra notes 1830, 1832, 1834, and 1835 and accompanying text for relevant details on these cost estimates.

a. Estimates for Rule 612

Each trading venue will have to update systems to comply with the change in tick size for some NMS stocks under the Rule 612 amendments. Due to similarities with the changes that were required by the TSP, the Commission estimated, in the Proposing Release, that the

amendments to Rule 612 would impose the same costs to trading venues as those estimated for the TSP.¹⁶³⁴ These costs were estimated at \$140,000 in 2014 at the time of the TSP.¹⁶³⁵

Some commenters stated that the \$140,000 estimated cost to trading venues in the proposal was too low because exchanges would have to acquire additional hardware and update various systems.¹⁶³⁶ These commenters did not provide alternative estimates for the implementation costs. As discussed in the Proposing Release, this \$140,000 estimate is derived from exchange feedback on the costs associated with the TSP.¹⁶³⁷ This estimate acknowledges that the market participants may have hardware and system costs associated with the amendments. Given that those hardware and system changes are similar in nature to those associated with the TSP, and the commenters did not provide analysis to the contrary, the Commission continues to believe that this estimate is reasonable. One commenter suggested that the costs of processing and disseminating trading information may increase linearly with any increases in message traffic.¹⁶³⁸ However, estimating the costs in this manner is not possible as the Commission does not know the current costs incurred by exchanges in processing and disseminating trading information, is unaware of data sources that could provide reliable estimates, and commenters did not provide such information. As discussed in section VII.D.1.c, the Commission acknowledges that message traffic may increase due to the amendments to Rule

¹⁶³⁴ See Proposing Release, supra note 11, at 80333.

¹⁶³⁵ An exchange commenting on the Tick Size Pilot estimated \$140,000 as its expected expense to comply with the Tick Size Pilot's requirement to change the tick size for some stocks. See James G. Ongena, Chicago Stock Exchange (CHX), Comment Letter Re: File No. 4-657; Notice of Filing of the Proposed National Market System Plan to Implement a Tick Size Pilot Program On a One-Year Pilot Basis (Dec. 2014), available at <https://www.sec.gov/comments/4-657/4657-67.pdf>.

¹⁶³⁶ See FIF Letter at 9-10 and FISD Letter at 3.

¹⁶³⁷ See Proposing Release, supra note 11, at 80333 n.618 and surrounding text.

¹⁶³⁸ See FIF Letter at 9 ("Some FIF members would estimate that the increased server, bandwidth and other costs would be roughly proportional to the increase in message traffic").

612, and so the costs of processing and disseminating message traffic may similarly increase. There is, however, uncertainty as to whether exchanges will need to incur additional hardware investments, and the degree of such investments, if they are needed, will likely differ from exchange to exchange, as it would depend on the capacity of their existing infrastructure to handle increased data.

To account for likely increases in the costs of computer hardware since 2014, the Commission is revising the estimated costs from \$140,000 to \$156,000 per trading venue.¹⁶³⁹ As shown in Table 15, the Commission estimates that the compliance costs associated with the amendments of Rule 612 across all trading venues are \$43 million. This estimate is computed by multiplying an estimated \$156,000 in one-time costs incurred by each trading venue to update systems to comply with the amendments to Rule 612, by the estimated number of trading venues, which is 277 trading venues.¹⁶⁴⁰

Under the amendments to Rule 612, listing exchanges will have to calculate NMS stocks' time-weighted average quoted spreads and transmit their associated tick size to the exclusive SIPs. The Commission does not believe the reduction in the number of tick sizes relative to the proposal will significantly affect compliance costs since the TWAQS still needs to be calculated

¹⁶³⁹ The Commission has updated the expected costs to \$156,000 from \$140,000 to reflect the roughly 11.4% increase in the Producer Price Index for Data Processing, Hosting and Related Services from December 2014, when the \$140,000 estimate was first made. See U.S. Bureau of Labor Statistics, Producer Price Index by Industry: Data Processing, Hosting and Related Services: Hosting, Active Server Pages (ASP), and Other Information Technology (IT) Infrastructure Provisioning Services [PCU5182105182105] (Mar 11, 2024), retrieved from FRED, Federal Reserve Bank of St. Louis, available at <https://fred.stlouisfed.org/series/PCU5182105182105>.

¹⁶⁴⁰ The technical aspect of a broker-dealer that internalizes customer orders updating its system to reflect the tiered tick regime is likely similar to that of an exchange or an ATS. Thus, the Commission is applying the same cost estimate for wholesalers and other broker-dealers that execute customer orders to update their systems as that applied to exchanges and ATSs. In Q1 2023 there were 16 registered exchanges, 33 ATSs, and 228 other FINRA members (including wholesalers) that executed orders off-exchange. In the first quarter of 2023, there were 277 total entities affected. See Rule 605 Amendments, supra note 10, at 26542.

for each stock and assigning a stock to a tick size is not computationally or conceptually difficult once the TWAQS has been computed. Thus, the Commission is keeping the estimate for listing exchanges the same at \$33,000 per listing exchange¹⁶⁴¹

Commenters stated that modifications to the data specifications with regards to transferring the tick size information from the listing exchanges to the SIPs would require both internal and external testing by the primary listing exchange and the SIPs.¹⁶⁴² Commenters stated that modifications to data specifications require software changes and require testing.¹⁶⁴³ The Commission anticipates that the SIPs will not have to acquire additional hardware or develop new systems in order to incorporate the minimum pricing increment indicator; they will rather need to update existing specifications. The Commission expects that the amendments to Rule 612 will require a one-time cost for updating existing systems and will not increase the cost of becoming a competing consolidator once the MDI Rules are implemented, because the amendments are not expected to increase the cost of establishing new systems. Accordingly, the

¹⁶⁴¹ The \$33,000 estimate per listing exchange is based on the following calculations: \$19,950 (hourly rate for Sr. Programmer at \$399 for 50 hours) + \$6,860 (hourly rate for Sr. Systems Analyst at \$343 for 20 hours) + \$3,730 (hourly rate for Compliance Manager at \$373 for 10 hours) + \$2,940 (hourly rate for Director of Compliance at \$588 for 5 hour) $\approx$ \$33,000. Salaries for estimates in this section are derived from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits and overhead.

¹⁶⁴² See generally CTA-UTP Operating Committee Letter and FIF Letter.

¹⁶⁴³ Id.

Commission estimates a one-time cost of \$13,000¹⁶⁴⁴ and ongoing costs of \$9,000 per year¹⁶⁴⁵ for the two SIPs.

These estimates are based on the Commission's understanding that the listing exchanges currently have access to the data needed to calculate the time-weighted average quoted spreads because such data, specifically the NBBO, are needed for the exchanges to compile Rule 605 reports.¹⁶⁴⁶ Thus, the Commission does not expect that the exchanges will incur additional costs associated with gathering data. Additionally, the listing exchanges have experience computing a share-weighted measure of average quoted spreads for their Rule 605 reports.¹⁶⁴⁷ The listing exchanges also already have connections to the exclusive SIPs, and once competing consolidators replace the exclusive SIPs it is the competing consolidators that will have the responsibility to connect to the exchanges in order to receive data. Thus, under the MDI Rules

¹⁶⁴⁴ The \$13,000 estimate per listing exchange is based on the following calculations: \$4,788 (hourly rate for Sr. Programmer at \$399 for 12 hours) + \$1,715 (hourly rate for Sr. Systems Analyst at \$343 for 5 hours) + \$3,730 (hourly rate for Compliance Manager at \$373 for 10 hours) + \$2,940 (hourly rate for Director of Compliance at \$588 for 5 hour) $\approx$ \$13,000. Salaries for estimates in this section are derived from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits and overhead.

¹⁶⁴⁵ The \$9,000 estimate per listing exchange is based on the following calculations: (\$2,640 (hourly rate for Compliance Attorney at \$440 for 6 hours) + \$746 (hourly rate for Compliance Manager at \$373 for 2 hours)) x 4 tick size revisions per year $\approx$ \$9,000.

¹⁶⁴⁶ See, e.g., 17 CFR 242.605(a)(ii)(E), requiring the reporting of the number of shares executed with price improvement, and 17 CFR 600(b)(36) defining "executed with price improvement" to mean, for buy orders, execution at a price lower than the national best offer at the time of order receipt and, for sell orders, execution at a price higher than the national best bid at the time of order receipt.

¹⁶⁴⁷ See, e.g., 17 CFR 242.605(a)(ii)(A), requiring the reporting of the average quoted spread for executions of covered orders, and 17 CFR 600(b)(12), defining the average quoted spread as the share-weighted average of the difference between the national best offer and the national best bid at the time of order receipt or, for order executions of midpoint-or-better limit orders, the difference between the national best offer and the national best bid at the time such orders first become executable. Additionally, some listing exchanges have issued white papers that include statistics based on time-weighted average quoted spreads. See, e.g., Nasdaq Intelligent Tick, supra note 150, Chart 3 and Cboe Proposal, supra note 150, Exhibit 1.

the exchanges will not incur additional costs to connect to the competing consolidators.¹⁶⁴⁸

Additionally, the SIPs have experience distributing regulatory data and so the costs represent those of adding the tick size to existing data. Consequently, the Commission expects that having the listing exchange compute time-weighted average quoted spreads and transmit the associated tick to the exclusive SIPs currently, or to the competing consolidators once the exclusive SIPs are discontinued, will require listing exchanges to modify existing systems, rather than build or acquire new systems or hardware.

The Commission expects that broker-dealers with order entry systems will need to modify their existing systems to comply with the tick size changes and will not need to acquire new hardware or develop new systems for this specific aspect of the adopted rule. In the Proposing Release, the Commission had estimated the cost of a broker-dealer system change at \$11,000.¹⁶⁴⁹ One commenter stated that for most firms a significant technological build will not be needed,¹⁶⁵⁰ other commenters stated that broker-dealers would have to undertake significant systems work, or acquire additional hardware, as a result of the amendments to Rule 612, specifically if there is a significant increase in message traffic.¹⁶⁵¹ The message traffic implications and costs are discussed in section VII.D.1.c. This section deals specifically with modifications to broker-dealer order entry systems.

One commenter stated that the \$11,000 estimate was too low because the proposed amendments to Rule 612 would necessitate additional expenses in order to update “their order

¹⁶⁴⁸ See MDI Adopting Release, supra note 10, at 18612 n.1133 and surrounding text. The costs for the competing consolidators to connect to the exchanges is accounted for in the MDI Rules and thus would not represent costs associated with this proposal.

¹⁶⁴⁹ See Proposing Release, supra note 11, at 80332-34.

¹⁶⁵⁰ See Apex Letter at 15.

¹⁶⁵¹ See FIF Letter at 9, TradeStation Letter at 6, Citigroup Letter at 2, and FISD Letter at 3.

management, execution management, customer trading, middle-office trade processing, reporting, settlement, surveillance and compliance systems.”¹⁶⁵² In the Proposing Release, when the Commission estimated the cost of a broker-dealer system change at \$11,000, it assumed that a single system change would be needed in response to the Rule 612 amendments.¹⁶⁵³ In light of the various functions described by the commenter which would need to be updated, the Commission is revising the estimated one-time cost to broker-dealers to \$33,000 per broker-dealer with an order-entry system as reported in Table 15.¹⁶⁵⁴ The revised estimate stems from the expectation that broker-dealers are more likely to have to undertake three system changes, rather than one,¹⁶⁵⁵ although the Commission recognizes that these costs could be greater if additional hardware investment is needed.

The Commission estimates that there are 1,161 broker-dealers with order entry systems.¹⁶⁵⁶ Thus, the Commission estimates that the amendments will lead to a one-time aggregate cost of around \$38.5 million (i.e., \$38.5 million $\approx$ \$33,000 x 1,161) across broker-dealers with order entry systems to update their systems to account for the new tick sizes.

The Commission expects that broker-dealers with smart order routers will also need to modify their existing systems to comply with the tick size changes and will not need to acquire new hardware or develop new systems. The Commission estimates a one-time cost of \$11,000 to

¹⁶⁵² See FIF Letter at 9.

¹⁶⁵³ See Proposing Release, supra note 11, at 80332.

¹⁶⁵⁴ See Table 15 note d.

¹⁶⁵⁵ The Commission believes that the order management, execution management, and customer trading functions highlighted by the commenter are sufficiently similar to be covered under a single system change. The middle-office trade processing, reporting, and settlement functions constitute another system change. Surveillance and compliance systems constitute a third system change.

¹⁶⁵⁶ Using CAT data from December 2023, the Commission calculated the total number of unique Central Registration Depository Numeric Identifiers “CRDs” that originated an order to estimate the number of entities with an order entry system.

broker-dealers operating smart order routers.¹⁶⁵⁷ These broker-dealers already have systems that can adjust for tick sizes that change around the \$1.00 threshold. Thus, the Commission expects that they will modify existing systems rather than build new systems. Any broker-dealer that will need to build new systems will likely incur costs greater than \$11,000 to do so. One commenter stated that significant system changes and hardware costs would also be required by broker-dealers operating smart order routers due to an increase in message traffic.¹⁶⁵⁸ The Commission acknowledges that the one-time costs to broker-dealers operating smart order routers may be greater than estimated if additional hardware investment will be required.¹⁶⁵⁹

The Commission estimates an upper bound of 270 broker-dealers operating smart order routers.¹⁶⁶⁰ This number provides an upper bound as it assumes that all entities with direct connections to exchanges or ATSs use a smart order router, which is likely an over-estimate. Aside from potential additional costs due to increased message traffic, the Commission thus estimates a one-time cost of \$3.0 million (i.e., \$3.0 million $\approx$ \$11,000 x 270) for market participants to update smart order routers.¹⁶⁶¹ If fewer than 270 broker-dealers operate their own

¹⁶⁵⁷ The \$11,000 estimate per broker-dealers with smart order routers is based on the following Manager at \$298 for 10 hours) + \$4,640 (hourly rate for Programmer Analyst at \$232 for 20 hours) + \$1,325 (hourly rate for Senior Business Analyst at \$265 for 5 hours) $\approx$ \$11,000. See also Securities Exchange Act Release No. 84875 (Dec 19, 2018), 84 FR 5202 (Feb 20, 2019) (“Transaction Fee Pilot Adopting Release”) at 5274 n.796 where the cost to broker-dealers to update systems for the TSP was estimated to be \$9,000. Here, we are allowing for an additional 10 hours of Programmer Analyst time.

¹⁶⁵⁸ See FIF Letter at 9.

¹⁶⁵⁹ See supra section VII.D.1.c for additional discussion about costs to market participants stemming from increases in message traffic.

¹⁶⁶⁰ This number is estimated by counting the number of unique CRDs that submitted an order directly to an exchange or ATS in the month of December 2023.

¹⁶⁶¹ The Commission also expects there may be other costs associated with updating systems to account for an increase in message traffic resulting from the new tick sizes. See supra section VII.D.1.c for additional discussion.

smart order routers, then the \$3.0 million estimate is likely higher than the aggregate cost for these broker-dealers to adjust their order routing systems to comply with these amendments.

Further, the Commission believes that broker-dealers operating smart order routers already subscribe to SIP data and will subscribe to consolidated market data products once the competing consolidators become operative. Thus, they will not incur a separate data expense to receive the regulatory messages necessary to comply with Rule 612 amendments. The Commission also assumes that system updates will impose a similar cost on larger and smaller entities given that, once code is written, scaling it up is relatively inexpensive.

Lastly, the Commission recognizes that Rule 612 amendments could increase the overall implementation costs of the MDI Rules. In particular, stocks that will become less tick-constrained as a result of the smaller tick size following these amendments could have more odd-lot quotes inside the NBBO than anticipated when the Commission adopted the MDI Rules.¹⁶⁶² As a result, the costs to SROs and competing consolidators of collecting, transmitting, consolidating, and disseminating odd-lot information will be greater than those described in the MDI Rules. The Commission is unable to provide an estimate of this cost because it would require predicting a complex interaction between behavior changes from multiple types of market participants and the resulting effect on the number of ticks inside the NBBO and the volume of odd-lots submitted inside the NBBO. However, the cost increase may not be significant, because the Commission generally estimates that the infrastructure cost increases associated with an increase in message traffic from the amendments to be approximately 1%.¹⁶⁶³

¹⁶⁶² This is a result of a smaller tick size allowing liquidity to spread over more levels, reducing the depth at each level and could increase the number of odd-lot quotes at each level. See supra section VII.D.1.b for additional discussion.

¹⁶⁶³ See supra note 1334 and accompanying text.